

SiPM-Wall MIP Calibration from Wall-Plastic Coincidences

n_TOF EAR2 X17 — run 224404 (first beam data, July 15)

Dylan Neff

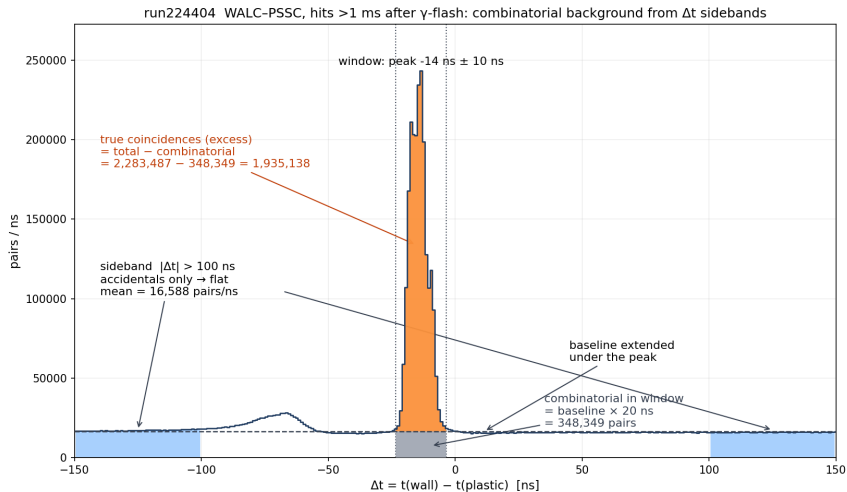
analysis and slides prepared with Claude (Anthropic AI assistant)

July 17, 2026

Data & selection

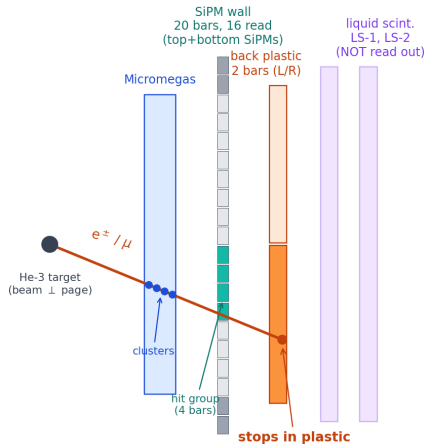
- ▶ Official processed file, run 224404: 3792 bunches (1471 dedicated), $\sim 300\text{M}$ hits.
- ▶ Per-channel ADC $\rightarrow$ mV from DAQ settings (≈ 0.0305 mV/count).
- ▶ SiPM walls delivered no data for 46% of beam-on bunches (643–2212) — plastics unaffected; hardware cause TBD with DAQ crew.
- ▶ All results: post-recovery bunches, hits > 0.1 ms after γ -flash (highest coincidence purity).

Method: coincidences with sideband subtraction

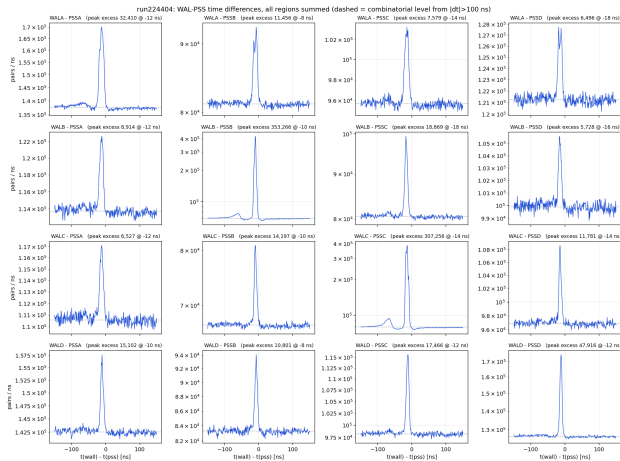


What the coincidence selects

Current selection: wall $\times$ plastic coincidence
(track needs only to REACH the plastic — MIP selection for the wall, hit selection for the plastic)

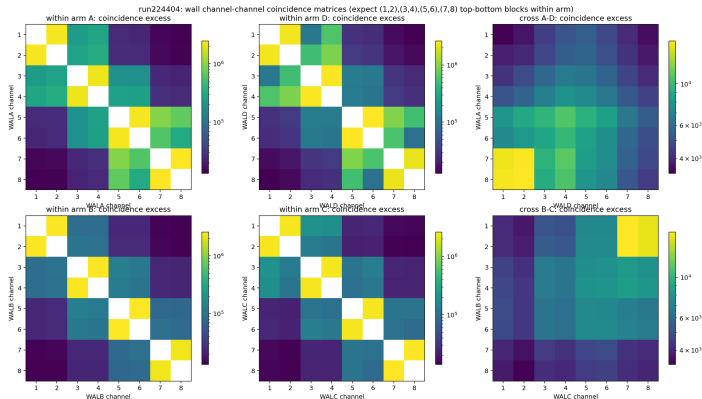


Every arm pairs wall $\leftrightarrow$ plastic cleanly



Peaks at -8 to -18 ns, rms 1.5–3.5 ns; per-channel offsets now a stored calibration.

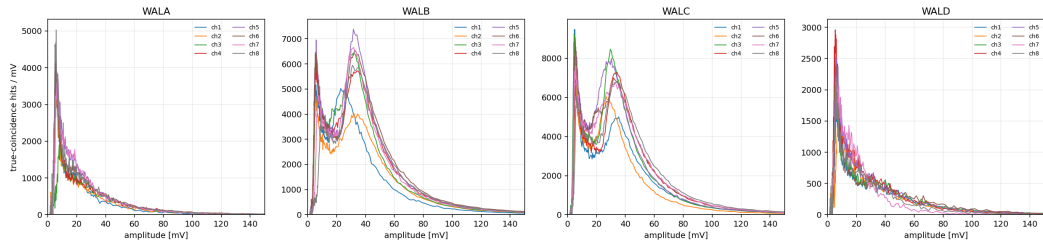
Geometry confirmed in data



Top/bottom blocks (1,2)(3,4)(5,6)(7,8) per 4-bar group; groups ordered left→right across the two bars. Corner hotspots $\Rightarrow$ A-D and B-C are *adjacent* arms. Odd=top is an assumption — need cabling sheet.

Headline: SiPM-wall MIP peaks on arms B and C

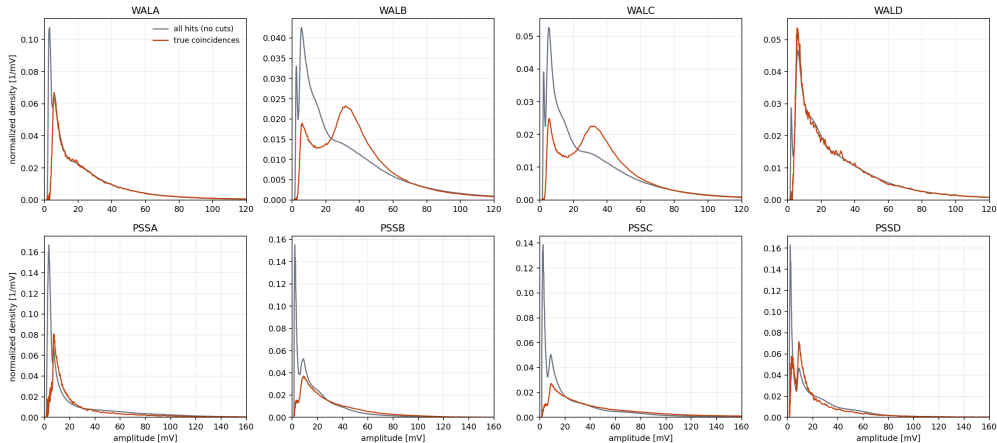
run224404: sideband-subtracted coincidence amplitude spectra (SiPM wall, tof > 0.1 ms, linear axes)



Sideband-subtracted coincidence spectra. **B/C: clean MIP peaks, 29–39 mV, all 16 channels $\Rightarrow$ per-channel calibration.** A/D: no MIP population (see later).

Coincidence selection performs exactly as intended (B/C)

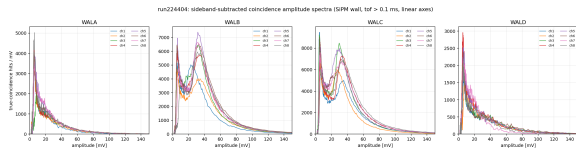
run224404: inclusive vs coincidence amplitude spectra, channels summed, unit area above 6 mV (linear)



Unit-normalized. Physics note: the plastic is at the *back* of the stack $\Rightarrow$ the coincidence is a through-going (MIP) selection *for the wall*, but only a hit selection for the plastic — plastic spectra are NOT MIP spectra.

The A/D problem — cause still open

- ▶ A/D: 6–8 $\times$ fewer true coincidences; coincident spectrum just traces the inclusive shape — no MIP bump.
- ▶ Hypotheses: (i) plastic response misses MIPs; (ii) wall SiPM gain low; (iii) geometry, absorption, or flux composition.
- ▶ Inclusive plastic medians agree within $\pm 30\%$ across all 8 PMTs (PSSD1 highest) $\Rightarrow$ large plastic-gain deficit unlikely.
- ▶ Wall-only top–bottom check (backup): SiPM response healthy on all four arms; but same-scintillator coincidence cannot distinguish through-going tracks from local deposits (e.g. γ /Compton), so the cause remains open.



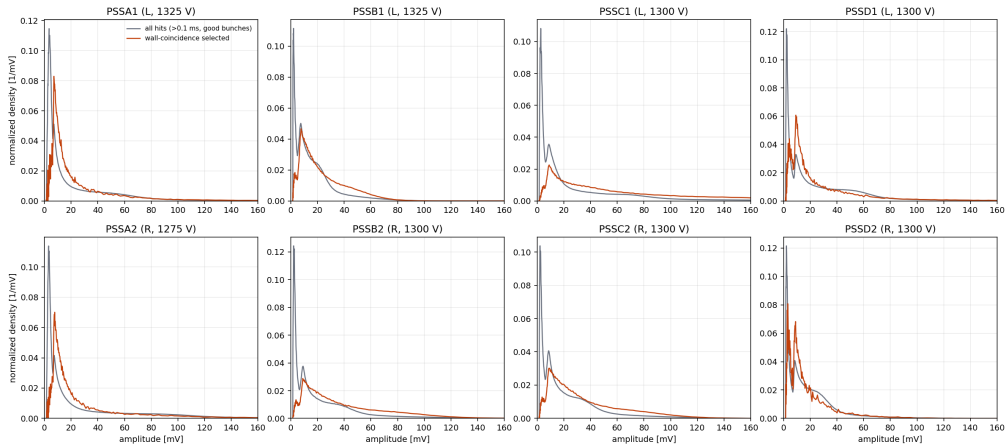
Plastic PMT relative gains & HV suggestions

PMT	HV [V]	med _{inc} [mV]	rel. gain	V(equalize)	V(2×)
PSSA1 (AL)	1325	10.4	0.74	1384	1528
PSSA2 (AR)	1275	13.2	0.93	1288	1422
PSSB1 (BL)	1325	13.1	0.93	1339	1478
PSSB2 (BR)	1300	14.9	1.05	1290	1425
PSSC1 (CL)	1300	14.5	1.03	1295	1429
PSSC2 (CR)	1300	15.3	1.08	1285	1419
PSSD1 (DL)	1300	17.8	1.26	1258	1389
PSSD2 (DR)	1300	14.9	1.06	1290	1424

$G \propto V^7$ assumed — request a 2-point HV scan to calibrate.

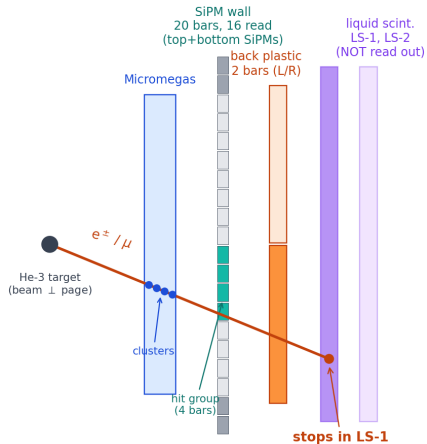
Raw vs coincidence-selected plastic spectra (per PMT)

run224404: plastic PMT spectra, raw vs wall-coincidence selected (unit area above ~ 2 mV)



What we want to enable later today: LIQ readout

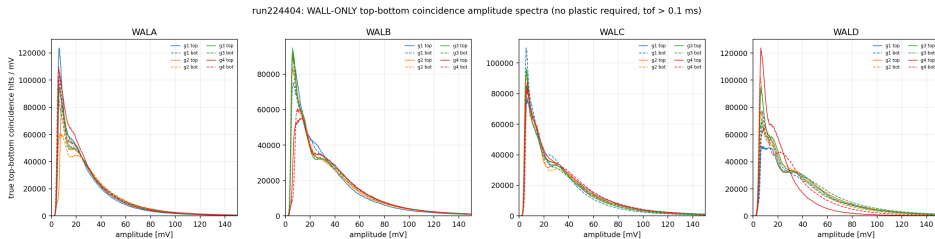
With LIQ readout (want today): wall $\times$ plastic $\times$ LS-1
(track must pass THROUGH the plastic — true MIP selection for the plastic)



Requests

1. **Read out the liquid scintillators** (behind the plastic): wall+LIQ coincidence requires passage *through* the plastic $\Rightarrow$ true plastic MIP calibration. Only LIQA/LIQD cabled, no data.
2. Plastic HV scan (2 points) $\rightarrow$ apply gain equalization.
3. Cabling confirmations: top/bottom card assignment; which 4 of 20 wall bars unread; physical arm positions A–D.
4. DAQ follow-up on the SiPM-wall outage (bunches 643–2212).

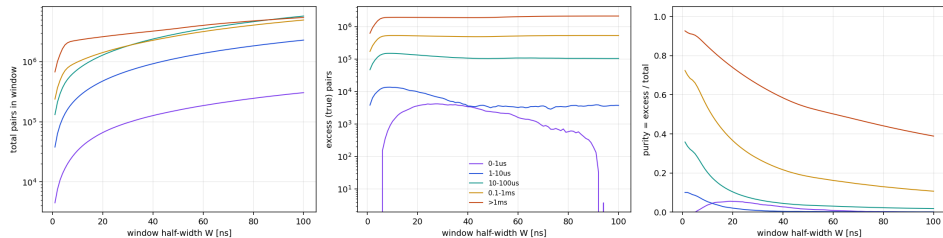
Backup: wall-only top-bottom coincidences



- ▶ Top & bottom SiPMs read the *same* 4-bar group: any real deposit (through-going MIP, stopping $e^\pm$, or a γ Compton-scattering in the bars) lights both ends in tight coincidence — removes noise, not physics backgrounds.
- ▶ Shows: SiPM response/gain healthy on all four arms (MIP-scale bump 25–40 mV everywhere); an amplitude landmark for all 32 channels.
- ▶ Cannot show: whether particles traverse onward to the back plastic — the A/D deficit remains unattributed (plastic response, absorption between wall and plastic, or flux composition).

Backup: window scan & purity

run224404: WALC-PSSC window scan by time since flash (peak at -14 ns)



Backup: full QA

